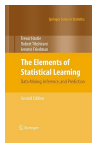


Lesson 02.3: ML Techniques - Input Processing

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References:

- Burkov: *"The Hundred-Page Machine Learning Book"* (2019)
- Russell et al.: *"Artificial Intelligence: A Modern Approach"* (4th ed, 2020)
- Hastie et al.: *"The Elements of Statistical Learning"* (2nd ed, 2009)



- *Input Processing*

Data Processing Transformations

- **Purpose of data processing**
 - Prepare raw data for effective machine learning
 - Improve model performance and generalization
- **Data cleanup**
 - Apply filters or smoothing to remove irrelevant variations
- **Handling missing data**
- **Types of transformations**
 - Normalization and standardization
 - Encoding categorical data
 - Feature construction
 - Dimensionality reduction
 - Discretization
- **Data augmentation**
 - Increase dataset size using transformations (common in vision)

Data Cleaning

- **Purpose of data cleaning**
 - Ensure data quality for accurate model training
 - Detect and correct errors or inconsistencies in the dataset
- **Typical steps in data cleaning**
 - *Remove duplicates*: Identical records are eliminated
 - *Correct data entry errors*: Fix misspellings or misformatted entries
 - *Standardize data*:
 - * Convert dates formatted as both MM/DD/YYYY and DD-MM-YYYY into consistent format
 - * String normalization (e.g., lowercase conversion)
 - * Type conversion (e.g., strings to integers)
 - * Dealing with unexpected characters or encodings
- **Relevance to ML models**
 - Poor data quality leads to biased or incorrect predictions
 - Clean data reduces variance and improves generalization

Handling Outliers and Missing Data

- **Outliers**

- *Definition*: Data points significantly different from others
- *Causes*: Measurement errors, variability in the data
- *Detection*: Box plots, Z-scores, or interquartile method
- *Treatment*: Removal, capping, or transformation (e.g., log scale)

- **Missing data**

- *Detection*: Count of null values or incomplete entries

- **Remediation**

- *Deletion*: Remove rows or columns with too many missing values
- *Imputation*:
 - * Mean/median/mode substitution
 - * K-nearest neighbors (KNN)
 - * Regression or model-based approaches
 - * E.g., for a missing temperature reading, impute using the mean of the day's surrounding values

Normalization and Standardization

- **Goal:** Adjust feature scales for better convergence and learning
 - *Normalize*: Rescale to $[0, 1]$
 - *Standardize*: Zero mean, unit variance
- **Why it helps**
 - Equal feature contribution in distance-based models
 - Faster convergence in gradient-based algorithms
 - Enables regularization
 - Easier feature interpretation
- **Common methods**
 - Min-Max normalization: $x' = \frac{x - x_{min}}{x_{max} - x_{min}}$
 - Z-score standardization: $x' = \frac{x - \mu}{\sigma}$

Encoding Categorical Data

- **Goal:** Convert non-numeric categories into numeric representations
- **Label encoding**
 - Assigns an integer to each category
 - * E.g., red, green, blue $\rightarrow$ 1, 2, 3
 - Can mislead models if order is not meaningful
- **One-hot encoding**
 - Creates binary vector per category
 - * E.g., red, green, blue $\rightarrow$ [1,0,0], [0,1,0], [0,0,1]
 - Avoids ordinal assumption
 - Increases dimensionality

Feature Construction

- **Goal:** Derive more informative features from raw inputs
- **Methods**
 - Combining variables (e.g., $\text{area} = \text{height} \times \text{width}$)
 - Extracting parts (e.g., year from a date)
 - Logical features
 - * E.g., transform 2023-04-15 $\rightarrow$ (Saturday, is_weekend = True)
- **Why it helps**
 - Encodes domain knowledge
 - Improves model expressiveness and performance

Dimensionality Reduction

- **Goal:** Reduce number of features while preserving key information
- **Why it helps**
 - Reduce overfitting
 - Reduce data redundancy
 - Improve model speed
 - Allow visualization
- **Common techniques**
 - PCA: linear combinations that maximize variance
 - LDA: projects data to maximize class separability
- **Example**
 - Reduce 1024x640 image pixels to 10 principal components
 - Quantize color images into gray scale

Discretization

- **Goal:** Convert continuous values into categorical bins
- **Why it helps**
 - Simplifies models or enables categorical algorithms
 - Helps detect threshold effects in data
- **Techniques**
 - Equal-width binning
 - Quantile binning
- **Example**
 - Discretize age
 - * **Child:** $[0, 13)$
 - * **Teen:** $[13, 20)$
 - * **Adult:** $[20, 65)$
 - * **Senior:** $[65, \infty)$
 - Age 32 $\rightarrow$ **Adult**

Noise Removal

- **Goal:** Remove irrelevant or corrupt data variations
- **Why it helps**
 - Improves signal clarity and model robustness
 - * E.g., clean noisy speech by removing high-frequency noise
- **Methods**
 - Smoothing (e.g., moving average)
 - Filtering (e.g., low-pass filter in audio)